

Technical Document #97: Deep Learning - Comprehensive Overview

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Generative adversarial networks (GANs) consist of two neural networks competing against each other. The generator creates synthetic data while the discriminator evaluates authenticity.

Dropout is a regularization technique where randomly selected neurons are ignored during training. This prevents co-adaptation and improves generalization.

Residual connections enable training of very deep neural networks by providing shortcuts for gradient flow. They were introduced in the ResNet architecture.

Transformer architecture relies entirely on self-attention mechanisms, dispensing with recurrence and convolutions. It has become the foundation for modern NLP models.

Transfer learning is a technique where a model trained on one task is re-purposed on a second related task. It has revolutionized computer vision and NLP.

Batch normalization normalizes the inputs of each layer, reducing internal covariate shift. It allows higher learning rates and acts as a regularizer.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Key Takeaways:

- Batch normalization normalizes the inputs of each layer, reducing internal covariate shift.
- Recurrent neural networks (RNNs) are designed to process sequential data.
- Transfer learning is a technique where a model trained on one task is re-purposed on a second related task.